ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE

PILANI (RAJASTHAN)

PortIQ – AI-Powered Portfolio Intelligence

Final Project Report

MBACCZG583 – Management of AI Products

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Table of Contents

Contents

1. Product Strategy and Opportunity Framing

Product Vision

Enable portfolio leaders to make faster, evidence-based commercial and delivery decisions by transforming portfolio data into AI-assisted business intelligence.

JTBD

When I am managing multiple client accounts and projects, I want to quickly understand portfolio health, identify risks and prioritise actions, so that I can improve profitability, delivery performance and customer outcomes.

Opportunity Framing

Portfolio information is fragmented across project management, ERP, CRM and reporting systems. Existing dashboards present data but require manual interpretation. Generic AI assistants lack portfolio-specific business context. PortIQ combines deterministic portfolio metrics with AI-generated explanations to support faster and more consistent management decisions.

AI Opportunity Matrix

Idea	Business Value	Feasibility	Priority
Portfolio Copilot	High	High	High
Predictive Risk Alerts	High	Medium	Medium
Resource Optimisation	Medium	Medium	Medium
Executive Narrative Reports	Medium	High	Medium

Build vs Buy vs Partner/API

Recommended path: Partner/API.
Reason: Use an enterprise LLM API while retaining proprietary portfolio logic, orchestration and business calculations within PortIQ.
Trade-offs: Faster delivery and lower infrastructure effort, balanced against API cost and vendor dependency.

Platform vs Feature

PortIQ is positioned as a standalone portfolio intelligence platform. The Copilot is a flagship AI feature within the platform rather than the product itself.

First Wedge

The initial use case is Portfolio Copilot for executive portfolio reviews. It enables managers to identify low-margin accounts, loss-making projects and portfolio risks through evidence-grounded AI responses.

2. PM Artifact Pack

Problem Statement

Portfolio leaders struggle to convert fragmented operational data into timely commercial decisions, resulting in delayed interventions and reduced portfolio profitability.

Product Vision

Create an AI-powered portfolio decision intelligence platform that augments managerial judgement through explainable, evidence-based recommendations.

PRD Summary

User Problem: Slow, manual portfolio analysis.
Core Workflow: Review dashboard → Ask Copilot → Receive evidence-backed insight → Take management action.
AI Role: Interpret validated portfolio metrics and generate business insights.
Risks: Hallucinations, outdated data and user over-reliance.
Fallback Logic: Display evidence, decline unsupported requests and keep human approval.

Feature Prioritisation

Feature	User Value	Effort	Confidence	Priority
Dashboard	High	Low	High	P1
Portfolio Analytics	High	Medium	High	P1
PortIQ Copilot	High	Medium	High	P1
Projects	Medium	Low	High	P2
Predictive Analytics	High	High	Medium	P3

MVP Definition

The MVP proves that portfolio leaders can use AI to obtain grounded business insights from portfolio data while maintaining transparency, explainability and human oversight.

Roadmap

Phase 1: MVP with Dashboard, Portfolio, Clients, Projects, Reports and Copilot.
Phase 2: Live enterprise integrations and role-based access.
Phase 3: Predictive analytics, forecasting and enterprise-wide deployment.

Success Metrics

Metric Type	Metric	Target
Product	Monthly active users	70% pilot adoption
Business	Portfolio review effort	50% reduction
Trust	Grounded responses	95% consistency with calculation engine

3. UX, Workflow, and Trust Design

Main Workflow

1. User reviews portfolio KPIs on the Dashboard.
2. User identifies an area requiring attention.
3. User asks PortIQ Copilot a business question.
4. AI orchestration layer retrieves relevant portfolio metrics.
5. Copilot returns an evidence-grounded response.
6. User validates the evidence and takes the final management decision.

Wireframes / Screens

Figure 1 – Executive Dashboard

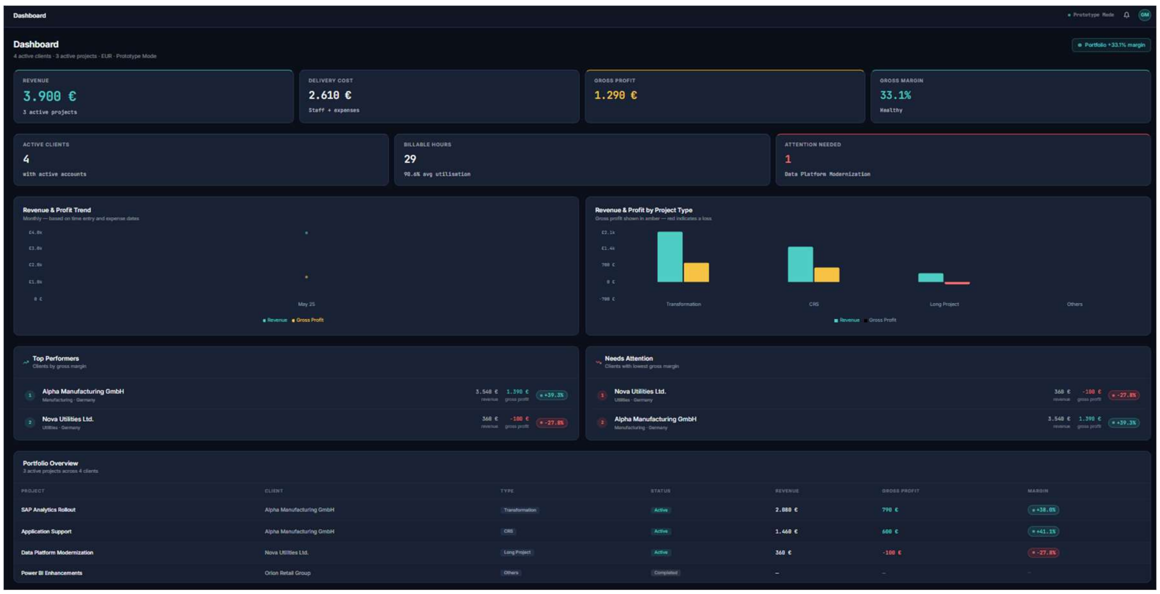


Figure 2 – Portfolio Analysis

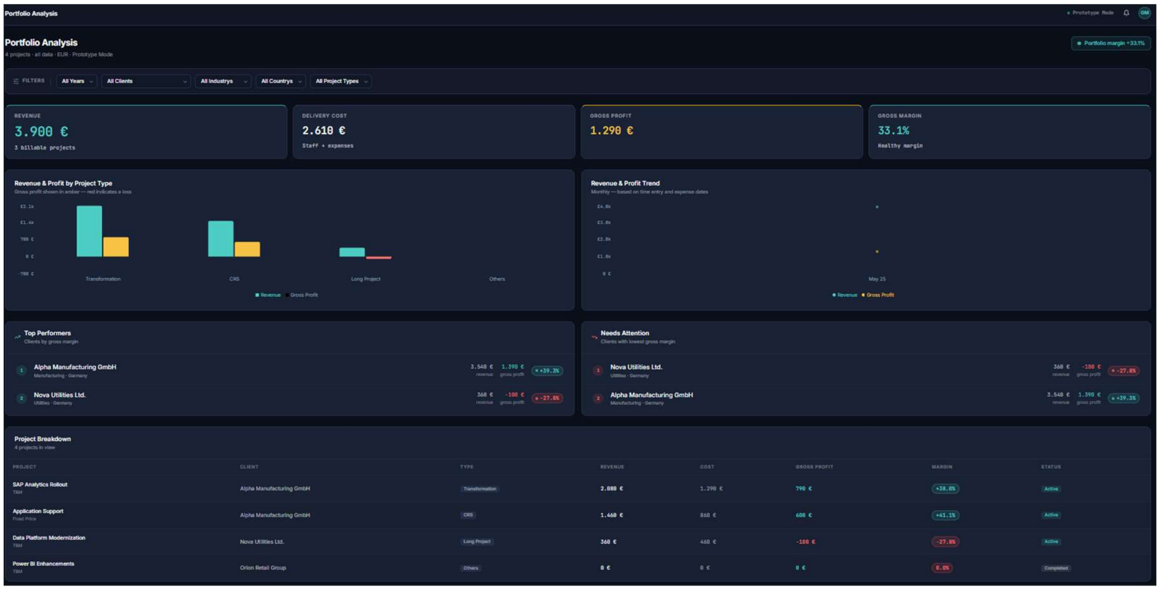


Figure 3 – Clients

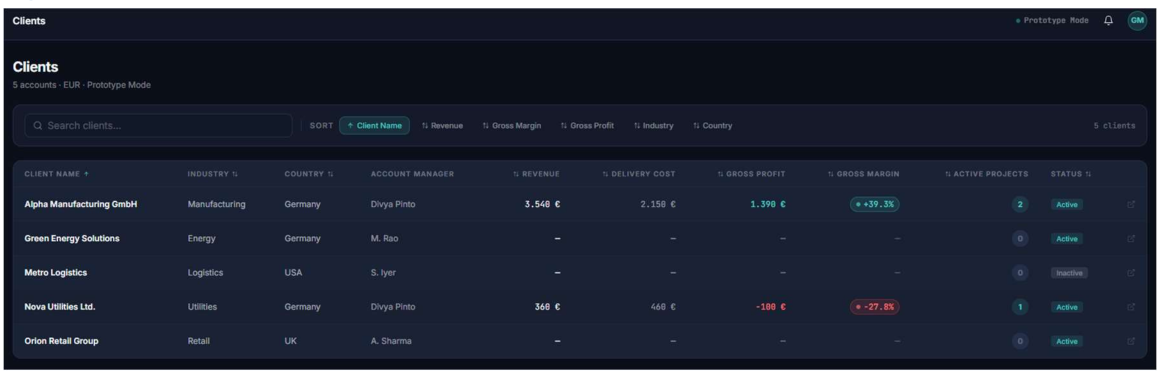


Figure 4 – Projects

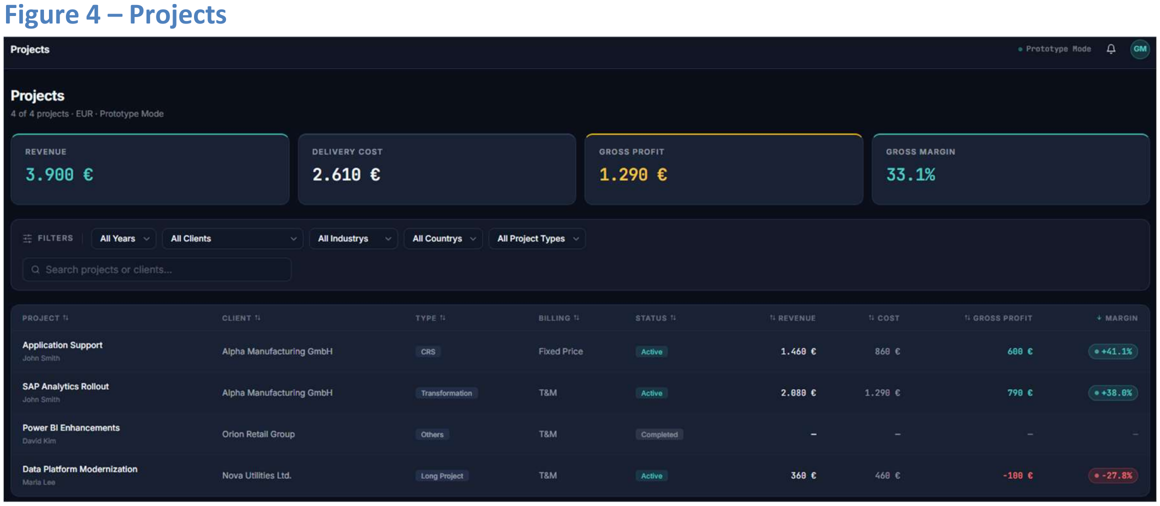


Figure 5 – Reports

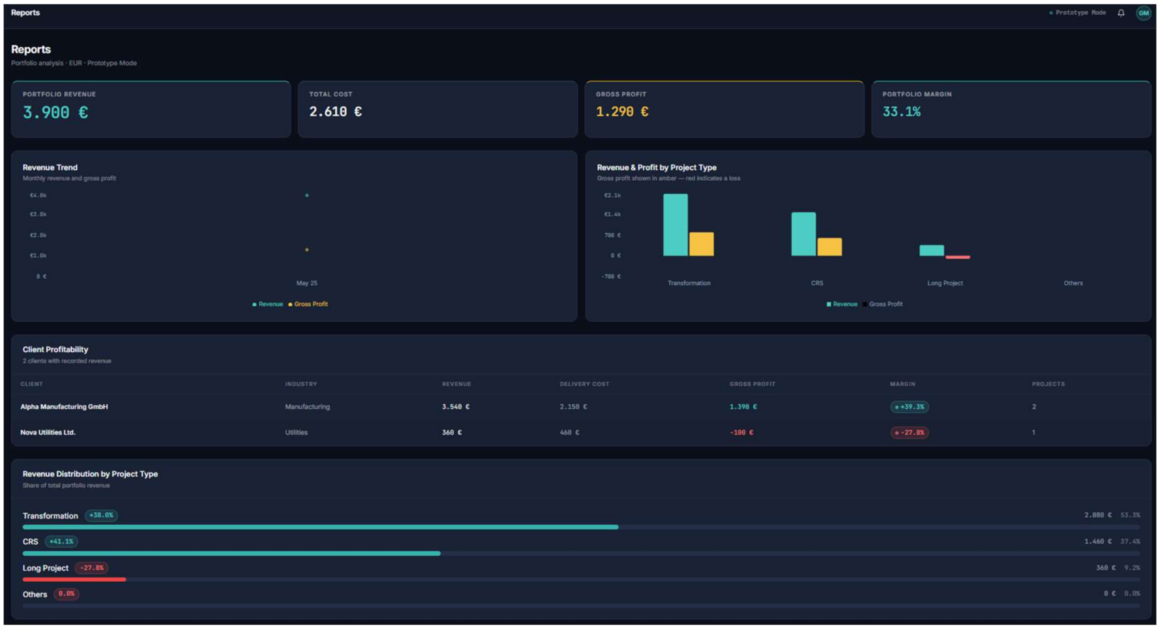
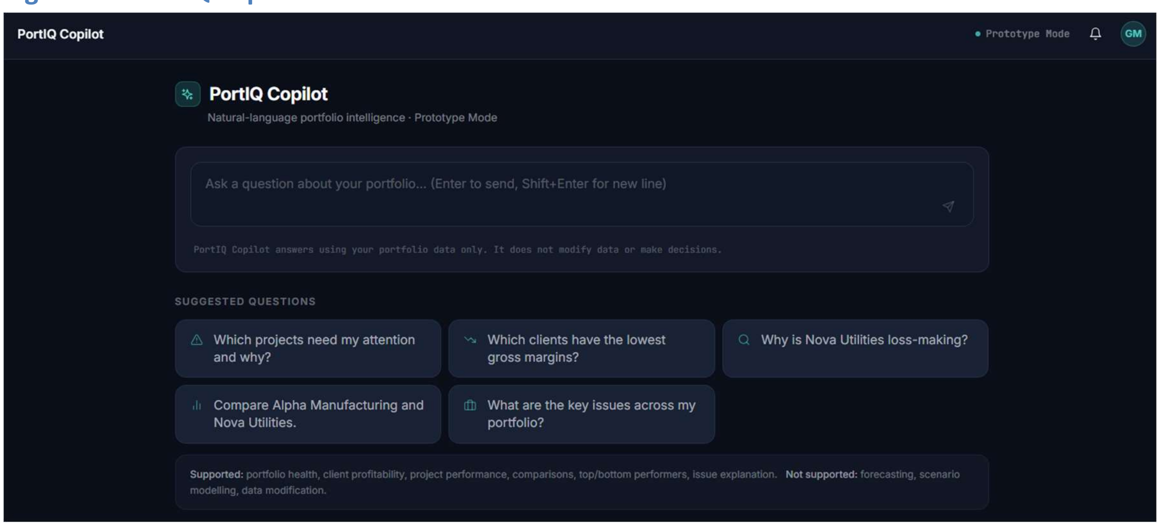


Figure 6 – PortIQ Copilot



Trust and Explainability

The user should always know that recommendations are generated using verified portfolio metrics. Supporting evidence, portfolio KPIs and calculation results should remain visible, while prompt construction and model implementation details remain hidden. The Copilot is presented as a decision-support assistant and not as an autonomous decision maker.

Uncertainty Handling

Scenario	Behaviour

High confidence	Return recommendation with supporting evidence
Low confidence	Advise user to review dashboard and available metrics
Failure / API unavailable	Display graceful fallback message and continue using dashboards

Human-in-the-Loop

Human review occurs before every commercial or operational decision. The AI provides analysis and recommendations, while portfolio managers remain accountable for approvals and actions.

Solution Architecture

The PortIQ architecture is presented at two levels. The MVP architecture illustrates the implementation developed for this academic prototype using synthetic data and a lightweight AI integration. The target-state enterprise architecture demonstrates how the solution can scale into a production environment by integrating enterprise data sources, an AI orchestration layer and governance controls.

Figure 7. PortIQ MVP Architecture

The MVP architecture consists of a React-based web application developed using Vite and Tailwind CSS. Portfolio metrics are computed through a deterministic calculation engine operating on a synthetic dataset. User queries are processed by the AI Copilot service, which constructs prompts using validated portfolio context before invoking the LLM through a secure serverless API. This architecture demonstrates the end-to-end AI-assisted decision workflow while remaining lightweight and suitable for rapid prototyping.

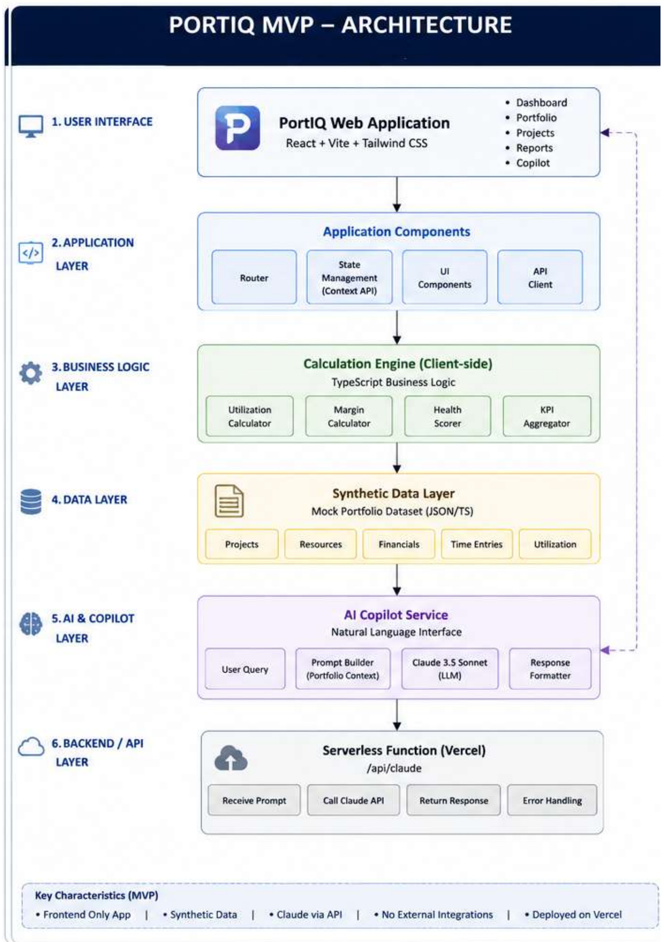
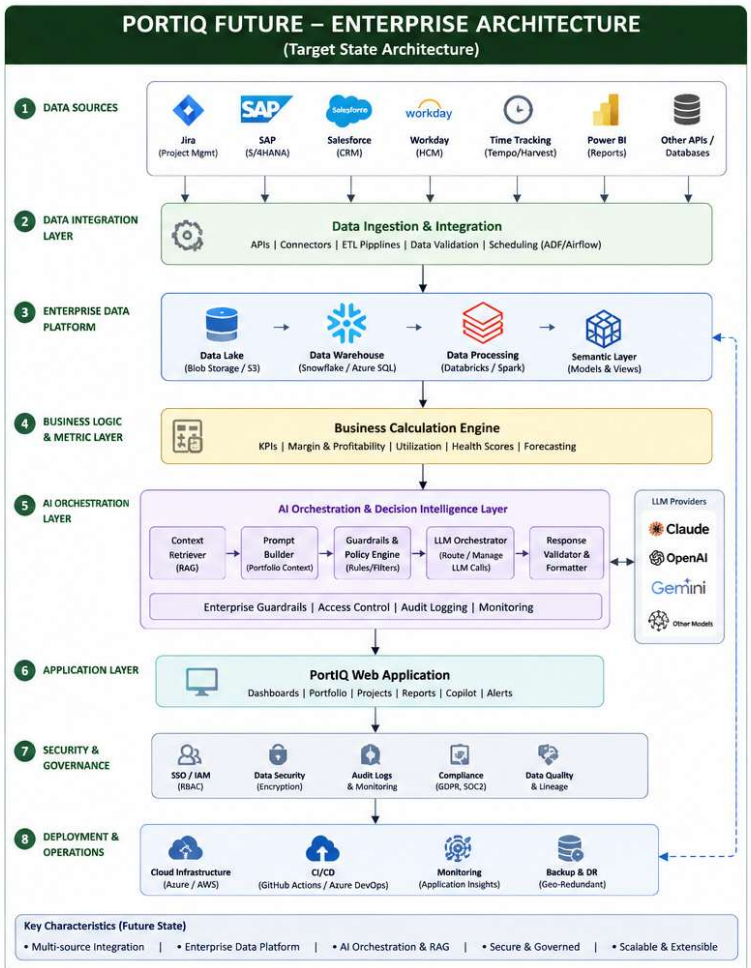


Figure 8. PortIQ Future Enterprise Architecture

The target-state enterprise architecture extends the MVP into a production-ready platform. Portfolio data is integrated from enterprise systems such as Jira, SAP, Salesforce, Workday and time-tracking applications through a data integration layer into a central enterprise data platform. Business KPIs are calculated by the calculation engine and supplied to the AI orchestration layer, which performs context retrieval, prompt construction, guardrail enforcement and LLM orchestration. Security, governance, monitoring and deployment services ensure scalability, compliance and operational reliability.



4. Data, Model, and Evaluation Strategy

Problem as AI Task

The solution is primarily a retrieval and generation task. The AI interprets verified portfolio metrics and generates business explanations and recommendations rather than predicting or modifying operational data.

Data Sources

Source	Examples
Internal	Jira, Azure DevOps, SAP ERP, CRM, HRIS, Time Tracking
External	Market benchmarks (future scope)
User-generated	Copilot questions and interaction history

Labeling / Ground Truth

Ground truth is provided by deterministic portfolio calculations including revenue, cost, gross margin, utilisation and project profitability. AI responses are expected to remain consistent with these validated metrics.

Baseline

The baseline alternative is conventional dashboard-based portfolio analysis without AI assistance.

Model / API Choice

A commercial LLM API is recommended together with an orchestration layer that retrieves portfolio context, applies guardrails and submits only validated business metrics to the model.

Evaluation Framework

Metric Type	Metric	Why It Matters
Technical	Grounded response accuracy	Ensures consistency with business calculations
Business	Reduction in portfolio review effort	Measures productivity improvement
Trust	User confidence and acceptance	Validates explainability and adoption

Launch Thresholds

The product should proceed beyond pilot if it demonstrates at least 70% user adoption during the pilot, reduces portfolio review effort by approximately 50%, maintains high consistency with portfolio calculations and receives positive feedback on recommendation quality.

5. Business, Economics, and Scaling

Monetization Model

PortIQ is positioned as an enterprise SaaS platform with an annual subscription model priced per organisation or business unit. For internal deployment, value is realised through productivity improvements, reduced portfolio review effort and improved commercial decision making.

Value Logic

The platform creates value by reducing manual portfolio analysis, identifying low-margin accounts earlier, improving resource allocation and enabling evidence-based executive decisions. AI augments managers rather than replacing them.

Cost Drivers

Cost Driver	Description
LLM/API	Inference requests
Cloud	Hosting and storage
Operations	Monitoring and maintenance
Human Review	Governance and model oversight

Unit Economics View

Healthy portfolio margins improve the business case for adoption. Recommended management thresholds are: >35% High Performing, 20–35% Healthy, 10–20% Watchlist, 0–10% Commercial Review and <0% Immediate Intervention.

Scaling Risks

Key risks include API cost growth, latency, governance complexity, enterprise integration effort and user adoption.

6. Ethics, Governance, and Risk Note

Main Risks

- Hallucinated recommendations
- Exposure of sensitive portfolio information
- User over-reliance on AI outputs

Harm Scenarios

Incorrect recommendations may influence commercial decisions, pricing or customer prioritisation if not reviewed by managers.

Controls

Ground AI responses using verified portfolio calculations, implement role-based access control, maintain audit logs, provide evidence with every recommendation and require human approval before major business decisions.

Launch Blockers

Production launch should be delayed if response quality is inconsistent, governance controls are absent or enterprise data security requirements are not satisfied.

7. Prototype / MVP

Prototype Type

The prototype is a React-based web application that demonstrates an AI-powered portfolio decision intelligence platform. It integrates interactive portfolio dashboards, business analytics and an AI-assisted PortIQ Copilot to support evidence-based portfolio management using synthetic business data.

What It Demonstrates

The prototype demonstrates the complete portfolio review workflow from data visualisation to AI-assisted decision support. Users can analyse portfolio KPIs, review client and project performance, explore management reports and interact with the PortIQ Copilot to obtain evidence-grounded business insights based on portfolio metrics. The MVP architecture and the target-state enterprise architecture are presented in Section to illustrate both the implemented solution and the proposed production deployment model.

AI Role

The AI interprets validated portfolio metrics retrieved through the AI orchestration layer and generates natural language explanations, portfolio comparisons and decision-support recommendations. All financial calculations are performed by the deterministic calculation engine, while the AI focuses solely on reasoning and explanation. Human managers remain responsible for all final business decisions.

What It Does Not Yet Prove

The MVP uses a synthetic portfolio dataset and does not include live enterprise integrations, production authentication, predictive analytics, forecasting or a production LLM deployment. These capabilities are identified as future phases of the product roadmap..

Links

Repository: https://github.com/divyaongit-lab/PortIQ-AI-Portfolio-Intelligence
Screenshots: Included in section 3
Setup:

1. npm install
2. npm run dev